

Equal Areas of Vertex Triangles in a Regular Polygon

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Abstract

Let $a(n)$ be the number of distinct areas of triangles whose vertices are vertices of a regular n -gon. We prove, for $n \geq 3$,

$$a(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor - \mathbf{1}_{4|n} \left\lfloor \frac{n-4}{8} \right\rfloor - \mathbf{1}_{24|n} - 2\mathbf{1}_{30|n}.$$

Thus all equal-area coincidences come from one rational one-parameter family, present when $4 \mid n$, together with one sporadic pattern of denominator 24 and two of denominator 30. The proof converts equality of areas into a conjugation-stable vanishing relation of twelve roots of unity and applies the classification of Poonen and Rubinstein; it is entirely computer-free.

1 Introduction

A triangle with vertices among those of a regular n -gon is determined up to congruence by the three positive arc lengths between its consecutive vertices. Hence the number of triangle shapes is simply the number of partitions of n into three positive parts,

$$p_3(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor.$$

The number of shapes is easy to count; the harder question is whether two noncongruent shapes can have the same area. Such coincidences do occur, but the answer turns out to be unexpectedly rigid: apart from a single infinite family tied to divisibility by 4, there are only three sporadic rational patterns.

The main idea is to rewrite equality of two area products as a vanishing relation among twelve roots of unity. Poonen and Rubinstein classified the minimal vanishing relations needed at this weight. The triangle constraints $u + v + w = x + y + z = 1$ translate into two additional phase conditions, which eliminate all but three decomposition types. Two of the surviving types give the three sporadic collisions; the third gives the one-parameter family.

Sections 2 and 3 translate equal-area conditions into vanishing sums of roots of unity and determine the three possible collision mechanisms. Section 4 counts the resulting coincidences and proves the formula for $a(n)$. The sequence $a(n)$ is OEIS A400213.

2 Triangles and their areas

Choose three vertices of a regular n -gon. Let r, s, t be the numbers of polygon sides in the three arcs between consecutive chosen vertices. Then

$$r, s, t > 0, \quad r + s + t = n.$$

The unordered triple (r, s, t) determines the triangle up to congruence. Hence the number of triangle shapes is

$$p_3(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor,$$

which is OEIS A069905.

If the circumradius is R , the side lengths are

$$2R \sin \frac{\pi r}{n}, \quad 2R \sin \frac{\pi s}{n}, \quad 2R \sin \frac{\pi t}{n},$$

and the area is

$$2R^2 \sin \frac{\pi r}{n} \sin \frac{\pi s}{n} \sin \frac{\pi t}{n}.$$

Put

$$u = \frac{r}{n}, \quad v = \frac{s}{n}, \quad w = \frac{t}{n}.$$

Two shapes have the same area if and only if

$$\sin(\pi u) \sin(\pi v) \sin(\pi w) = \sin(\pi x) \sin(\pi y) \sin(\pi z), \quad (1)$$

where all six numbers are positive rationals and

$$u + v + w = x + y + z = 1. \quad (2)$$

Example 2.1 (The first collision). For $n = 12$, the two shapes

$$(1, 4, 7) \quad \text{and} \quad (2, 2, 8)$$

are not congruent, since their arc partitions differ. Their normalized area products nevertheless agree:

$$\sin \frac{\pi}{12} \sin \frac{\pi}{3} \sin \frac{7\pi}{12} = \frac{\sqrt{3}}{8} = \sin^2 \frac{\pi}{6} \sin \frac{2\pi}{3}.$$

Thus the regular dodecagon already has fewer distinct triangle areas than triangle shapes. Theorem 4.1 will show that this is the first such coincidence and that $a(12) = 11$.

Since $A + B + C = \pi$ implies

$$4 \sin A \sin B \sin C = \sin(2A) + \sin(2B) + \sin(2C),$$

Equation (1) is equivalent to

$$\sin(2\pi u) + \sin(2\pi v) + \sin(2\pi w) = \sin(2\pi x) + \sin(2\pi y) + \sin(2\pi z).$$

Define

$$\begin{aligned} \alpha_1 &= \frac{1}{2} - 2u, & \alpha_2 &= \frac{1}{2} - 2v, & \alpha_3 &= \frac{1}{2} - 2w, \\ \alpha_4 &= \frac{3}{2} - 2x, & \alpha_5 &= \frac{3}{2} - 2y, & \alpha_6 &= \frac{3}{2} - 2z. \end{aligned}$$

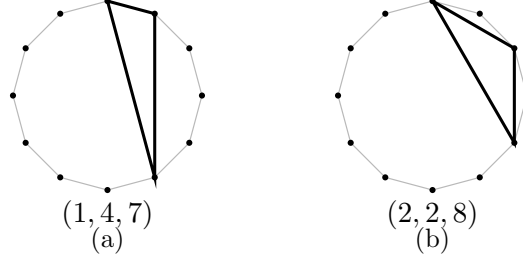


Figure 1: The first equal-area collision, in the regular 12-gon. The labels give the three cyclic arc gaps between consecutive chosen vertices. The two triangles are noncongruent but have the same area.

Then

$$\sum_{j=1}^6 e^{i\pi\alpha_j} + \sum_{j=1}^6 e^{-i\pi\alpha_j} = 0. \quad (3)$$

Moreover, (2) gives the two separate phase conditions

$$\alpha_1 + \alpha_2 + \alpha_3 = -\frac{1}{2}, \quad \alpha_4 + \alpha_5 + \alpha_6 = \frac{5}{2}. \quad (4)$$

Thus (3) is a conjugation-stable vanishing relation of weight 12 among roots of unity, subject to substantially stronger conditions than conjugation stability alone.

3 The three collision mechanisms

For a prime p , write

$$R_p : \quad 1 + \zeta_p + \zeta_p^2 + \cdots + \zeta_p^{p-1} = 0,$$

where $\zeta_p = e^{2\pi i/p}$. The *weight* of a relation is its number of terms, counted with multiplicity. Thus, for example, $4R_3$ denotes a decomposition into four rotated three-term relations. A relation is *conjugation-stable* if it is unchanged by $z \mapsto \bar{z}$.

Poonen and Rubinstein [1] classify, up to rotation, the minimal vanishing relations of roots of unity of weight at most 12; see their Theorem 3. Their Table 2 lists the 23 possible decomposition types of a relation of weight 12. Appendix A combines this classification with (4).

Proposition 3.1. *If two noncongruent positive rational triples of sum 1 satisfy (1), then the associated relation (3) has one of the three decomposition types*

$$4R_3, \quad 2R_3 + 3R_2, \quad 6R_2.$$

The first two types yield exactly the following three solutions, up to permutation within each triple and interchange of the two triples:

$$\left(\frac{1}{12}, \frac{5}{12}, \frac{1}{2}\right) \sim \left(\frac{1}{8}, \frac{1}{4}, \frac{5}{8}\right), \quad (5)$$

$$\left(\frac{1}{30}, \frac{11}{30}, \frac{3}{5}\right) \sim \left(\frac{1}{15}, \frac{1}{5}, \frac{11}{15}\right), \quad (6)$$

$$\left(\frac{2}{15}, \frac{2}{5}, \frac{7}{15}\right) \sim \left(\frac{1}{5}, \frac{7}{30}, \frac{17}{30}\right). \quad (7)$$

Here $\sim$ denotes equality of areas. The derivations are given in Appendices B and C. Figure 2 shows the unique denominator-24 pattern.

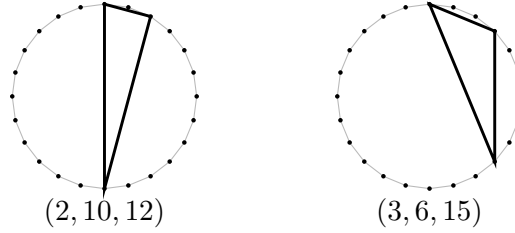


Figure 2: The unique sporadic collision of denominator 24. The right-hand shape is the fixed shape $t = 1/8$ of the one-parameter family, but the pair itself is an additional isolated collision.

The type $6R_2$ is qualitatively different and gives an infinite family. We treat it directly because it explains the main term in the correction to $p_3(n)$.

Lemma 3.2. *The nontrivial positive solutions arising from the type $6R_2$ are exactly*

$$\left(t, \frac{1}{2} - 2t, \frac{1}{2} + t\right) \sim \left(2t, \frac{1}{4} - t, \frac{3}{4} - t\right), \quad 0 < t < \frac{1}{4}, \quad (8)$$

up to the usual symmetries. At $t = 1/8$ the two unordered triples coincide.

Proof. In a decomposition of type $6R_2$, the twelve roots in (3) can be paired into six opposite pairs. For a fraction on the first side and one on the second side, an opposite-root pairing gives

$$x = u, \quad u + x = \frac{1}{2}, \quad \text{or} \quad u + x = \frac{3}{2}.$$

For two fractions on the same side, a nonfixed pairing forces them to differ by $1/2$; the alternative that they sum to 1 is impossible because the third fraction is positive.

Let a be the number of same-side pairs on each side, c the number of cross-pairs, and f the number of fixed fractions $1/2$ on each side. Counting the three positions on either side gives

$$2a + c + f = 3, \quad f \in \{0, 1\}.$$

There are only four possibilities for (a, c, f) . If $(a, c, f) = (0, 3, 0)$, all pairs are cross-pairs; the relations $x = u$, $u + x = 1/2$, and $u + x = 3/2$, together with the two triple sums, only fix or interchange the three fractions, so the unordered triples are equal. If $(a, c, f) = (0, 2, 1)$, both triples contain $1/2$ and the remaining two fractions sum to $1/2$; the two cross-pairs again only fix or interchange them. The case $(1, 0, 1)$ is impossible because a same-side pair already has sum greater than $1/2$. Thus a nontrivial solution can occur only for $(a, c, f) = (1, 1, 0)$: one same-side pair on each side and one cross-pair.

Write the same-side pairs as

$$t, t + \frac{1}{2} \quad \text{and} \quad s, s + \frac{1}{2}.$$

The remaining fractions are $1/2 - 2t$ and $1/2 - 2s$, with $0 < t, s < 1/4$. Their cross-pairing is either trivial, giving $s = t$, or their sum is $1/2$, giving

$$s = \frac{1}{4} - t.$$

The latter case is exactly (8). At $t = 1/8$ the two unordered triples are the same. $\square$

The equality in (8) can also be checked directly, which makes the family less mysterious. Its two normalized area products are

$$\sin(\pi t) \cos(2\pi t) \cos(\pi t) = \frac{1}{4} \sin(4\pi t)$$

and

$$\sin(2\pi t) \sin\left(\frac{\pi}{4} - \pi t\right) \sin\left(\frac{3\pi}{4} - \pi t\right) = \frac{1}{4} \sin(4\pi t).$$

4 Counting the collisions

For (8) to occur in a regular n -gon, write $t = k/n$. The terms containing $1/4$ force $4 \mid n$. Write $n = 4m$. Then (8) becomes

$$(k, 2m - 2k, 2m + k) \sim (2k, m - k, 3m - k), \quad 1 \leq k < m.$$

The parameters k and $m - k$ give the same pair, and $k = m/2$ is the trivial fixed point when it is integral. Hence the family contributes

$$\left\lfloor \frac{m-1}{2} \right\rfloor = \left\lfloor \frac{n-4}{8} \right\rfloor$$

nontrivial pairs when $4 \mid n$.

The solution (5) occurs exactly when $24 \mid n$ and gives, for $n = 24h$,

$$(2h, 10h, 12h) \sim (3h, 6h, 15h).$$

The solutions (6)–(7) occur exactly when $30 \mid n$ and give, for $n = 30h$,

$$(h, 11h, 18h) \sim (2h, 6h, 22h)$$

and

$$(4h, 12h, 14h) \sim (6h, 7h, 17h).$$

The pairs are distinct. The 24-solution contains the fixed shape $(1/8, 1/4, 5/8)$ of (8), but its other shape is not in that family; neither 30-solution belongs to the family. Proposition 3.1 and Appendices B–C also rule out a larger equal-area class: otherwise two shapes belonging to different listed pairs would themselves form an additional collision. Thus every listed nontrivial collision reduces the number of distinct areas by exactly one.

Theorem 4.1. *For $n \geq 3$,*

$$a(n) = \left\lfloor \frac{n^2 + 6}{12} \right\rfloor - \mathbf{1}_{4 \mid n} \left\lfloor \frac{n-4}{8} \right\rfloor - \mathbf{1}_{24 \mid n} - 2\mathbf{1}_{30 \mid n}.$$

Table 1 shows the smallest values of n for which the three correction mechanisms appear:

In particular, two noncongruent vertex triangles of a regular n -gon have equal area if and only if

$$(4 \mid n \text{ and } n \geq 12) \quad \text{or} \quad 30 \mid n.$$

n	$p_3(n)$	4-family	24-pattern	30-patterns	$a(n)$
12	12	1	0	0	11
24	48	2	1	0	45
30	75	0	0	2	73

Table 1: How the three collision mechanisms first contribute to the count.

A The phase reduction

This appendix proves Proposition 3.1. Every finite exclusion is supplied with an explicit residue certificate below.

A.1 Conjugation-stable decompositions

Poonen and Rubinstein's Lemma 4 states that a conjugation-stable relation of weight at most 12 can be decomposed into minimal relations so that every minimal component is either conjugation-stable or paired with its conjugate. Their proof uses a finite Mathematica check only in the final exceptional case $2(R_5 : R_3)$. We replace that check by the following short argument.

Lemma A.1. *The exceptional type $2(R_5 : R_3)$ also admits such a conjugation-stable decomposition without a case enumeration.*

Proof. Take the standard relation

$$B = \{\zeta_6, \zeta_6^{-1}, \zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4\}.$$

Its power sums satisfy

$$\sum_{\beta \in B} \beta^2 = -2, \quad \sum_{\beta \in B} \beta^3 = -3, \quad \sum_{\beta \in B} \beta^5 = 5.$$

Suppose that

$$S = qB \uplus rB, \quad |q| = |r| = 1,$$

is stable under complex conjugation. Every power sum of S is real, hence

$$q^2 + r^2, \quad q^3 + r^3, \quad q^5 + r^5$$

are real. Write

$$q = e^{i\theta}, \quad r = e^{i\phi}, \quad s = \theta + \phi, \quad d = \theta - \phi.$$

The second-power condition gives

$$\sin s \cos d = 0.$$

If $s \equiv 0 \pmod{2\pi}$, then $r = \bar{q}$. If $s \equiv \pi \pmod{2\pi}$, the third- and fifth-power conditions give

$$\cos \frac{3d}{2} = \cos \frac{5d}{2} = 0,$$

so $d \equiv \pi \pmod{2\pi}$ and $\{q, r\} = \{1, -1\}$; each component is then conjugation-stable.

If instead $\cos d = 0$, then neither $\cos(3d/2)$ nor $\cos(5d/2)$ vanishes. The third- and fifth-power conditions therefore imply

$$\sin \frac{3s}{2} = \sin \frac{5s}{2} = 0,$$

whence $s \equiv 0 \pmod{2\pi}$ and again $r = \bar{q}$. This is exactly the required conclusion. $\square$

We may therefore use the conjugation-stable decomposition of Poonen and Rubinstein.

A.2 The phase equations

All exponents may be considered modulo 2. From (4), if L is the set of three exponents belonging to the first triangle, every admissible relation satisfies

$$\sum_{j=1}^6 \alpha_j \equiv 0 \pmod{2}, \quad \sum_{j \in L} \alpha_j \equiv \frac{3}{2} \pmod{2}. \quad (9)$$

Equivalently, the product of the six chosen roots is 1 and the product of the three chosen roots on the first side is $-i$.

For completeness, we describe the finite reduction explicitly. Represent a root $e^{2\pi i x}$ by $x \in \mathbb{Q}/\mathbb{Z}$. Let

$$E_p = \left\{ 0, \frac{1}{p}, \dots, \frac{p-1}{p} \right\}$$

represent R_p . If a root a of a relation S is replaced by a rotated relation T sharing the root b , the exponents of the resulting relation are obtained by replacing a with

$$\left\{ x + a - b + \frac{1}{2} : x \in T, x \neq b \right\} \pmod{1}. \quad (10)$$

This is the exponent form of the notation $(S : T)$ used by Poonen and Rubinstein. Repeated use of (10) gives the representatives in their Table 1.

For each of the first 22 decomposition types in their Table 2, Lemma A.1 reduces conjugation stability to self-conjugate components and conjugate pairs. Choosing one exponent from each conjugate pair leaves at most two rotation parameters. (The last type, $6R_2$, can carry more parameters and is treated directly in Lemma 3.2.) The two conditions in (9) are therefore just two linear congruences over $\mathbb{Q}/2\mathbb{Z}$.

Rather than recording only a list of failed cases, we give an explicit arithmetic certificate for every excluded type. Put

$$T = \sum_{j=1}^6 \alpha_j, \quad L = \sum_{j \in L} \alpha_j \quad \text{in } \mathbb{Q}/2\mathbb{Z}.$$

The required phases are

$$T = 0, \quad L = \frac{3}{2}. \quad (11)$$

For a fixed decomposition type, formula (10) gives the six chosen exponents as affine functions of at most one free rotation parameter in the cases that occur below. If such a parameter is present, the equation $T = 0$ first eliminates it. What remains is the residue obstruction in the last column of Table 2.

The notation in the last column is literal. For example, “ $21T$ odd” means that $21T$ is an odd integer modulo 42 for every allowed conjugation-stable realization and every choice of one representative from each conjugate pair. This is incompatible with $T = 0$ in $\mathbb{Q}/2\mathbb{Z}$. Likewise, if $T = 0$ and $L \in (1/15)\mathbb{Z}/2\mathbb{Z}$, then $L \neq 3/2$, since $15(3/2)$ is not an integer.

Here are representative checks illustrating all three kinds of certificate in the table.

For row 1, take a self-conjugate representative of $R_7 : 5R_3$. After doubling the root exponents to obtain the α -coordinates and multiplying by 21, its six conjugate pairs can be written

$$(2, 40), (4, 38), (10, 32), (14, 28), (15, 27), (16, 26) \pmod{42}.$$

No.	decomposition type	certificate excluding (11)
1	$R_7 : 5R_3$	$21T$ is odd.
2	$R_7 : R_5, 2R_3$	$105T \not\equiv 0 \pmod{5}$.
3	$R_7 : (R_5 : R_3), R_3$	no conjugation-stable realization (Poonen–Rubinstein, Cor. 1).
4	$R_7 : (R_5 : 2R_3)$	$105T$ is odd.
5	$R_{11} : R_3$	$33T$ is odd.
6	$(R_7 : 3R_3) + R_2$	$42T$ is odd.
7	$(R_7 : R_5) + R_2$	$70T$ is odd.
8	$(R_5 : 4R_3) + R_3$	if $T = 0$, then $L \in (1/15)\mathbb{Z}/2\mathbb{Z}$.
9	$(R_7 : 2R_3) + R_3$	$21T = 2k$ with $k \not\equiv 0 \pmod{21}$.
10	$(R_5 : 3R_3) + 2R_2$	if $T = 0$, then $L \in (1/15)\mathbb{Z}/2\mathbb{Z}$.
11	$(R_7 : R_3) + 2R_2$	if $T = 0$, then $L \in (1/21)\mathbb{Z}/2\mathbb{Z}$.
12	$(R_5 : 2R_3) + R_5$	if $T = 0$, then $L \in (1/15)\mathbb{Z}/2\mathbb{Z}$.
13	$R_7 + R_5$	$35T \not\equiv 0 \pmod{5}$.
14	$(R_5 : 2R_3) + R_3 + R_2$	$30T \not\equiv 0 \pmod{15}$.
15	$R_7 + R_3 + R_2$	$42T \not\equiv 0 \pmod{3}$.
16	$2(R_5 : R_3)$	if $T = 0$, then $30L \not\equiv 0 \pmod{15}$.
17	$(R_5 : R_3) + 2R_3$	if $T = 0$, then $L \in (1/45)\mathbb{Z}/2\mathbb{Z}$.
18	$(R_5 : R_3) + 3R_2$	if $T = 0$, then $60L \not\equiv 0 \pmod{5}$.
19	$2R_5 + R_2$	survives the phase test; treated below.
20	$R_5 + R_3 + 2R_2$	if $T = 0$, then $30L \not\equiv 0 \pmod{5}$.

Table 2: Explicit residue certificates for the first twenty weight-12 decomposition types. Each certificate follows directly from the representatives in Poonen–Rubinstein Table 1 and the substitution rule (10).

Choosing one entry from each pair always gives an odd total, because exactly one pair has odd entries. The other self-conjugate rotation has five odd pairs and one even pair, again giving an odd total; the other minimal variants have the same parity pattern. Hence $21T$ is always odd.

Rows 8, 10, 11, 12, and 17 illustrate the lattice obstruction. In rows 8, 10, and 12 all fixed offsets lie in $(1/15)\mathbb{Z}$, and the equation $T = 0$ keeps every allowed rotation in that same lattice, so $L \in (1/15)\mathbb{Z}/2\mathbb{Z}$. Row 11 is identical with 21 in place of 15. In row 17 the total equation is of the form

$$3q \in (1/15)\mathbb{Z}/2\mathbb{Z},$$

so $q \in (1/45)\mathbb{Z}/2\mathbb{Z}$ and hence $L \in (1/45)\mathbb{Z}/2\mathbb{Z}$. None of these odd-denominator lattices contains $3/2$.

Finally consider the free-rotation residue obstruction, represented by row 16. Writing the two conjugate $R_5 : R_3$ blocks with rotation q , substitution of the standard exponent set

$$\left\{ 0, \frac{1}{30}, \frac{7}{30}, \frac{13}{30}, \frac{19}{30}, \frac{2}{3} \right\}$$

into $T = 0$ determines q modulo the kernel of the total coefficient. Substitution into the left sum gives

$$30L \equiv 1, 2, \dots, 14 \pmod{15},$$

never 0. Since $30(3/2) = 45 \equiv 0 \pmod{15}$, the required left phase is impossible. The same one-line elimination gives rows 18 and 20 with modulus 5. Rows 2, 13, 14, and 15 are the corresponding total-phase versions. Thus every exclusion in Table 2 is certified by a displayed residue condition and requires no positivity test or machine enumeration.

It follows that among the first twenty types only $2R_5 + R_2$ survives the phase equations. Types 21–23 are respectively $4R_3$, $2R_3 + 3R_2$, and $6R_2$, and are treated separately below and in the main text.

A.3 Eliminating $2R_5 + R_2$

Solving the two phase congruences for the type $2R_5 + R_2$ leaves, up to permutation within each side, the following eight pairs of triples of α -classes modulo 2:

L	R	
$(1/10, 1/2, 9/10)$	$(3/10, 7/10, 3/2)$	
$(1/10, 1/2, 9/10)$	$(13/10, 3/2, 17/10)$	
$(1/2, 11/10, 19/10)$	$(3/10, 7/10, 3/2)$	
$(1/2, 11/10, 19/10)$	$(13/10, 3/2, 17/10)$	
$(1/2, 13/10, 17/10)$	$(1/10, 9/10, 3/2)$	
$(1/2, 13/10, 17/10)$	$(11/10, 3/2, 19/10)$	
$(3/10, 1/2, 7/10)$	$(1/10, 9/10, 3/2)$	
$(3/10, 1/2, 7/10)$	$(11/10, 3/2, 19/10)$	

(A.4)

This is the complete phase-compatible list for the type, obtained by inserting the two self-conjugate R_5 rotations and the conjugate-pair rotations into (9).

Every row contains the class

$$\alpha \equiv \frac{1}{2} \pmod{2}$$

on the left and the class

$$\alpha \equiv \frac{3}{2} \pmod{2}$$

on the right. But positivity requires

$$-\frac{3}{2} < \alpha < \frac{1}{2} \quad \text{on the left,} \quad -\frac{1}{2} < \alpha < \frac{3}{2} \quad \text{on the right.}$$

The forced classes occur only at the excluded endpoints. Equivalently, they give a zero arc fraction. Thus $2R_5 + R_2$ has no positive solution. Together with Table 2, this proves Proposition 3.1.

B The type $4R_3$

Throughout Appendices B and C, solutions are considered up to permutation within each triple, complex conjugation, and interchange of the two triples.

There are two self-conjugate rotations of R_3 in exponent coordinates,

$$A = \{0, 1/3, 2/3\}, \quad B = \{1/6, 1/2, 5/6\},$$

and we write P_3 for a conjugate pair of non-self-conjugate R_3 components. Four components therefore have only nine conjugation-stable patterns:

$$4A, 3A + B, 2A + 2B, A + 3B, 4B, 2A + P_3, A + B + P_3, 2B + P_3, 2P_3.$$

The phase check for all nine patterns is small enough to display completely:

pattern	consequence of $T = 0$ and the conjugate-pair condition	
$4A$	$L \in \{0, 2/3, 4/3\}$	
$3A + B$	a fixed conjugate class has odd multiplicity	
$2A + 2B$	$T \in \{1/3, 1, 5/3\}$	
$A + 3B$	a fixed conjugate class has odd multiplicity	
$4B$	$L \in \{1/3, 1, 5/3\}$	(B.1)
$2A + P_3$	$L \in (2/9)\mathbb{Z}/2\mathbb{Z}$	
$A + B + P_3$	a fixed conjugate class has odd multiplicity	
$2B + P_3$	$L \in 1/9 + (2/9)\mathbb{Z}/2\mathbb{Z}$	
$2P_3$	survives.	

None of the first eight rows can have $L = 3/2$. Thus, under these symmetries, a relation of type $4R_3$ relevant to a nontrivial collision consists of two conjugate pairs of R_3 components. Let $q, r \in [0, 2)$ be their rotation parameters in the α -coordinate. The only nontrivial phase pattern has left-hand representatives

$$q, \quad q + \frac{2}{3}, \quad -r - \frac{4}{3} \pmod{2}$$

and right-hand representatives

$$q + \frac{4}{3}, \quad r, \quad r + \frac{2}{3} \pmod{2}.$$

Using the unique lifts to the positivity intervals, the left sum in (4) gives

$$2q - r = \frac{13}{6}. \quad (12)$$

The right-hand lifts have exactly two admissible possibilities:

$$q + 2r = \frac{5}{2} \quad \text{or} \quad q + 2r = \frac{9}{2}. \quad (13)$$

The intervening boundary choice would put an exponent at an endpoint of a positivity interval and hence give a zero arc.

Combining (12) with the first equation in (13) gives

$$q = \frac{41}{30}, \quad r = \frac{17}{30}.$$

The lifted exponents are

$$-\frac{19}{30}, \frac{1}{30}, \frac{1}{10} \quad \text{on the left,}$$

and

$$\frac{7}{10}, \frac{17}{30}, \frac{37}{30} \quad \text{on the right.}$$

Converting back by

$$u = \frac{1/2 - \alpha}{2}, \quad x = \frac{3/2 - \alpha}{2},$$

gives

$$\left(\frac{1}{5}, \frac{7}{30}, \frac{17}{30}\right) \sim \left(\frac{2}{15}, \frac{2}{5}, \frac{7}{15}\right),$$

which is (7).

The second equation in (13) gives

$$q = \frac{53}{30}, \quad r = \frac{41}{30}.$$

The lifted exponents are

$$-\frac{7}{30}, \frac{13}{30}, -\frac{7}{10} \quad \text{on the left,}$$

and

$$\frac{11}{10}, \frac{41}{30}, \frac{1}{30} \quad \text{on the right.}$$

Hence

$$\left(\frac{1}{30}, \frac{11}{30}, \frac{3}{5}\right) \sim \left(\frac{1}{15}, \frac{1}{5}, \frac{11}{15}\right),$$

which is (6). These are therefore the only nontrivial positive solutions of type $4R_3$.

C The type $2R_3 + 3R_2$

For R_3 use the two self-conjugate rotations A, B defined in Appendix B, and write P_3 for a conjugate pair. For R_2 put

$$C = \{0, 1/2\}, \quad D = \{1/4, 3/4\},$$

and write P_2 for a conjugate pair of non-self-conjugate R_2 components. There are $4 \cdot 6 = 24$ conjugation-stable component patterns. The complete phase check is

	$3C$	$2C + D$	$C + 2D$	$3D$	$C + P_2$	$D + P_2$
AA	—	—	—	—	—	✓
AB	—	—	✓	—	✓	—
BB	—	—	—	—	—	✓
P_3	—	✓	—	✓	—	✓

(C.1)

Here a dash means that the two phase equations are inconsistent already in $\mathbb{Q}/2\mathbb{Z}$; a check mark means that the phase equations have a solution and positivity must still be imposed. Thus exactly seven patterns remain. Their positive outcomes are

component pattern	positive outcome
$AA + D + P_2$	the solution (5)
$AB + C + P_2$	trivial equal triples only
$AB + C + 2D$	none
$BB + D + P_2$	the solution (5)
$P_3 + 2C + D$	none
$P_3 + 3D$	none
$P_3 + D + P_2$	trivial or the solution (5)

The three occurrences of the nontrivial solution are different decompositions of the same root-of-unity relation. It is therefore enough to derive one representative. It is enough to take the left-hand exponents to be

$$0, \quad q, \quad q + 1 \pmod{2},$$

and the right-hand exponents are

$$\frac{1}{2}, \quad \frac{2}{3}, \quad \frac{4}{3}.$$

The right-hand sum is already $5/2$. The left phase condition is

$$0 + q + (q + 1) \equiv -\frac{1}{2} \pmod{2}.$$

Up to replacing q by $q + 1$, which only interchanges the two conjugate representatives, the phase equation gives

$$q = \frac{1}{4}.$$

The corresponding lifted exponents are

$$0, \quad \frac{1}{4}, \quad -\frac{3}{4} \quad \text{on the left,}$$

and

$$\frac{1}{2}, \quad \frac{2}{3}, \quad \frac{4}{3} \quad \text{on the right.}$$

Therefore

$$\left(\frac{1}{8}, \frac{1}{4}, \frac{5}{8}\right) \sim \left(\frac{1}{12}, \frac{5}{12}, \frac{1}{2}\right),$$

which is (5). The table above records the remaining six phase-compatible component patterns: three have no positive lift, one gives only identical unordered triples, and two give the same relation just derived (with $P_3 + D + P_2$ also admitting a trivial lift). Hence there is no second nontrivial positive solution of this type.

References

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